Project Details

Low-cost 3D-Printed RoboCup-Style Robot Benchmarking Against TurtleBot v2

Abstract

The RoboCup Small Size League (SSL) motivates fast, agile, and low-latency mobile robots operating under tight mechanical and control constraints. This project will design and build a low-cost differential-drive robot inspired by SSL requirements using 3D-printed structural components and readily available off-the-shelf electronics (microcontroller or SBC, motor drivers, DC motors with encoders, battery management, wireless communication). The robot will be integrated with a ROS-compatible control stack so that it can accept standard motion commands, publish odometry, and support repeatable performance testing. The core aim is twofold: produce a custom robot suitable for participation in the Robocup Small Size League, and quantitatively compare the custom robot to a TurtleBot v2 across a set of robotics benchmarks relevant to navigation and embodied autonomy. Benchmarks will include command latency, maximum and sustainable speed, acceleration and stopping distance, turning radius, manoeuvrability on tight paths, etc.

Resources

- https://scholar.sun.ac.za/bitstreams/7f61b61e-de62-4c7d-be5d-b5a4f2ca7ee6/download
- https://dspace.mit.edu/bitstream/handle/1721.1/137217.2/1609.04745.pdf?isAllowed=y&sequence=4
- https://ssl.robocup.org/wp-content/uploads/2025/04/2025_TDP_First-Order.pdf